CENTER FOR GEOSPATIAL SOLUTIONS
Lincoln Institute of Land Policy

WMO

Follow-On Peer-to-Peer Technical Workshop: Piloting Interoperable Hydrometeorological Data Systems for Multi-Hazard Early Warning Systems (Addis Ababa, September 14–17, 2026)

1. Key Takeaways

1. Interoperability Across the Hydrometeorological Data Chain

- **Effective early warning depends on reliable data flows from observation to decision-making**. Hydrometeorological data must move from monitoring stations through transmission, data management, quality control, forecasting, and ultimately the production of warnings and advisories. Gaps or bottlenecks anywhere along this chain can undermine the usefulness of the information.
- **Connecting existing systems is more important than adopting a single platform**. National institutions already use different systems for collecting, managing, and sharing data, including CLIMSOFT, Aquarius, HydroServer, and wis2box. Interoperability allows these systems to exchange information and support common operational workflows without requiring institutions to replace their existing infrastructure.
- **Open standards and consistent metadata** make these connections possible. Common data structures, standardized interfaces, and well-documented metadata allow information to be discovered, exchanged, interpreted, and used across different systems and institutions.
- **WMO's Hydrological Observing System (WHOS)** illustrates how interoperability can work across institutions. By connecting distributed hydrological data sources, WHOS enables broader data discovery and access while allowing institutions to retain responsibility for managing their own data.

2. New technology should reduce—not add to—fragmentation

- NMHS institutions already operate multiple proprietary and open-source systems, often introduced through different projects. **New tools need to complement and connect with what already exists rather than create additional silos.**
- As one participant rightly asked, **"Are we moving in one direction [toward interoperability], or are we scattering ourselves across too many systems?"**
- This means evaluating the **end-to-end data flow and institutional ecosystem**, rather than evaluating individual software packages in isolation.

3. Moving from demonstration to operations requires confronting real implementation challenges

- Hands-on work has brought up practical issues around **deployment**, **hosting, APIs, automated ingestion, timestamps, data formats, quality control, integration, storage, and existing infrastructure**.
- NBI's migration from DataSight to HydroServer illustrated the value of a **deliberate, phased transition**: testing and understanding the system first, mapping the existing data model to the



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new data model, migrating historical data, and then transitioning live data from real-time sensors and manual stations.

- **Open-source platforms can significantly reduce licensing costs and vendor dependence, while increasing flexibility**. NBI estimated that consolidating its regional water-resources data in HydroServer would reduce annual system costs from approximately $57,000 to $2,000—a 96% reduction, or about $55,000 in annual savings that could be redirected to priorities such as capacity building.

4. An open-source software is about people as much as code and technology

- An open-source software solution depends on developers, users, technical support, documentation, training, and communities willing to share expertise. Open-source software needs a growing community around it willing to keep growing and building a good base of knowledge.
- Open-source systems also **evolve through use**. User feedback can help improve functionality. For example, we identified a few opportunities for improving Hydroserver :
    - Data from the Mara and Kanzenze stations, in two different countries, showed missing timestamps at the same time, suggesting a possible connectivity or data transmission issue
    - HydroServer needs stronger functionality to detect and fill data gaps.
    - Delays in the synchronization of GEOGLOWS forecasts can temporarily prevent the GEOGLOWS API from retrieving streamflow forecasts.
- Ultimately, institutions remain responsible for **data quality that should start with a good maintenance of the sensors, hosting, maintenance, staff capacity, and operational ownership**.
- WMO's experience with ClimWeb demonstrates the importance of sustained mentoring and technical assistance alongside software development. ClimWeb is an example of formidable progress in Africa in number of alerts issued and stations reporting data to WMO.

5. National ownership and regional collaboration need to reinforce each other

- **Regional collaboration is essential for early warning**, particularly for transboundary hazards and shared river basins. Regional platforms, forecasting, shared data products, and peer learning can help YOU address common challenges.
- At the same time, **national institutions need to retain ownership** of their data, systems, workflows, thresholds, and decisions about what is shared.
- The goal is therefore not a single regional or technological solution, but **nationally owned systems that can interoperate effectively with other national, regional, and global systems**.
- **Pilot refinement strengthened collaboration between hydrological and meteorological services** within each country, aligning data-sharing workflows to support operational early warning systems.

2. Next Steps

2.1. Capture, Synthesize, and Share Workshop Outcomes

CGS Leads

Timeline: short term; ~4-week horizon

- We will produce a detailed workshop report capturing country use cases, demonstrated tools, dialogue outputs, and the next steps we've agreed on.
    - This report will iidentify cross-cutting themes including shared technical successes, technical gaps, institutional constraints, and opportunities for regional alignment to guide follow-on support.
    - Highlight alignment with WMO, NBI and related activities and projects
- Share the report and snapshots with participants and circulate to NOAA, NBI, ICPAC, CIAT and WMO peers so that insights benefit the wider region.

2.2. Technical Assistance

CGS Leads

Timeline: short-medium term; ~ 4-month horizon

- In collaboration with TT-W2FH, wis2box and HydroServer developers, and WMO secretariat, CGS commits CGS software development capacity to organize and deliver actual upstream contributions to wis2Box and HydroServer v2, code changes, bug fixes, and features scoped from country feedback, so that improvements land in the shared codebases and benefit every NMHS, not just the countries directly engaged.
- We will maintain a public backlog of issues, feature requests, and pull requests linking each upstream contribution to the country's need that motivated it, making the adoption-to-improvement loop visible to funders and peers. This public backlog will include known/existing issues in wis2box and hydroserver
- We will provide support to help countries build on the training they received in Addis Ababa, deepen their understanding of WIS2Box and HydroServer, and become more familiar with the features and capabilities of both systems.
- We will provide technical assistance to support pilot implementation and the development and testing of additional capacity-building materials.

2.3. Ongoing Coordination and Peer Exchange

WMO leads

Timeline: medium-long term; ~ 1+ year horizon

- Establish regular follow-up (e.g., technical check-ins, working sessions)
- Facilitate exchange across countries to share progress and lessons
- Coordinate partner support based on identified country needs, aligning efforts with regional initiatives such as the Nile Basin Initiative's implementation of HydroServer v2, WHOS, and ClimWeb.

Closing remarks

Distinguished Directors, colleagues, partners, and friends,

On behalf of the Lincoln Institute of Land Policy's Center for Geospatial Solutions and the organizing team, I want to thank the Government of Ethiopia for hosting us here in Addis Ababa and for the warm welcome we have received throughout the week.

Thank you to the Directors and teams from Meteorology and Hydrology for welcoming us to your institutions and giving us the opportunity to see your work firsthand. Those visits were an important reminder that the discussions we have had this week are ultimately about putting these systems to work in real operational settings.

Thank you as well to our U.S. Embassy representatives, particularly Mr. Yeneneh Teka, for being with us throughout the workshop, and to our partners and guest speakers from WMO, the Nile Basin Initiative, and ICPAC for contributing your experience and perspectives.

I would like to extend a special thank you to our trainers—Washington, Sara, Emily, and Peter—for all the work you put into organizing and delivering these sessions. Peter, we are especially grateful to you, for jumping in at the last minute and leading the wis2box training. The grace and commitment to education and knowledge exchange are deeply valued and appreciated. As a token of our gratitude, we would like to present you with this certificate of recognition for your support with training this week.

I would like to thank Ms Juliet Nakasagga for helping is plan and deliver this workshop, for keeping all of us informed and organized, and for serving as the master of ceremonies this week. Thank you for your faith in our team and our mission, and for going above and beyond.

I also want to recognize Ms. Belay and the team from Capital Hotel and Spa for taking such good care of us throughout the week.

As we close, I hope we leave Addis Ababa not simply with more technical knowledge, but with stronger connections across institutions and countries and a clearer sense of what comes next.

Building sustainable, interoperable systems requires more than good tools. It requires collaboration, capacity, and institutional ownership. The real value of these four days will be in what we are able to take forward together.

Thank you for your time, your expertise, and your willingness to engage so openly with each other this this week. Safe travels home, and I look forward to continuing the work together.